

How GPT Works – One Page Overview

What is GPT?

GPT (Generative Pre-trained Transformer) is a large language model that predicts the next token (word or sub-word) in a sequence. It is trained on massive text datasets to learn language patterns, reasoning, coding, and general knowledge.

1. Tokenization

Input text is split into tokens that are converted into numeric IDs.

2. Embeddings

Each token is transformed into a dense vector representing semantic meaning. Positional embeddings preserve word order.

3. Transformer Layers

Multiple Transformer blocks process the sequence. The key mechanism is **self-attention**, allowing every token to focus on the most relevant previous tokens. Feed-forward networks refine these representations.

4. Next-Token Prediction

The model outputs probabilities for the next token. The selected token is appended to the input, and the process repeats until completion.

Training

Pre-training teaches general language by predicting missing/next tokens. Fine-tuning and alignment improve helpfulness, safety, and instruction following.

Inference Pipeline

Prompt → Tokenizer → Embeddings → Transformer Layers → Probability Distribution → Next Token → Repeat → Final Response.

Key Concepts

- Context Window: Maximum tokens processed at once.
- Attention: Identifies important relationships between tokens.
- Temperature: Controls randomness of output.
- Top-k/Top-p: Sampling strategies for token selection.

Limitations

GPT can generate incorrect information (hallucinations), has context limits, and relies on training data plus reasoning during inference rather than a database of facts.